

Preliminary Comparison of Research Software Engineering and Data Science Competences

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Abstract: Research Software Engineers (RSEs) and Data Scientists (DSs) occupy overlapping yet distinct roles within the landscape of digitally skilled professionals. Both roles are highly software-focused and operate across a wide range of research domains, yet their communities and competency frameworks have evolved independently. This paper explores the intersections and distinctions between RSE and DS competencies, particularly as they relate to different phases of the research cycle.

Keywords: Research Software-Engineering, Data Science, Competences

1 Introduction

Computers and software have become vital elements of the research process across almost all domains. They enable researchers to collect and process ever-increasing amounts of data, simulate a wide range of phenomena on previously unexplored scales, and discover previously inconceivably complex structures in nature and societies via machine learning (ML). This importance of computation and digitally-aided data analysis in research means that digital skills are now required by researchers at all career levels. However, as there is a much wider range of digital skills required to support the modern research lifecycle than an individual researcher can master to perfection, there is increasing demand for people in specialist technical roles who can interface between researchers and their digital tools and processes. These technical professionals (increasingly becoming known within the UK research community as “digital Research Technical Professionals” – dRTPs), provide targeted help, support and specialist technical expertise to researchers. Below we introduce some of the technical roles that have emerged in the research landscape over recent years (roughly in the order in which the terms started appearing).

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- **Research Data Managers** (also known as Data Stewards) support the handling of data throughout the research data life-cycle including data acquisition, processing, storage and publication. The efficiency, validity and recognition of research in many domains today hinges on the quality, availability and reproducibility of data and data-transforming methods. Research Data Management (RDM) therefore has become a cross-cutting field with a large number of sub-topics that range from legal themes, e.g. licenses, data usage agreements and data protection laws over technical themes related to the organization, storage, transport, transformation, annotation and analysis of digital data to topics traditionally associated with libraries, such as the preservation, publication and dissemination of research data [<https://doi.org/10.31263/voebm.v74i1.6255>; <https://doi.org/10.5281/zenodo.4320504>]. Research Data Managers work closely with domain scientists but also facilitate researchers' communication with other services, such as the legal and IT department as well as the library. Within Germany, numerous regional data management networks have been formed. More information can be found at [fdminfo25].
- **Data Scientists** are professionals who combine expertise in a range of areas that may include programming, statistics, machine learning, and specialist research domain knowledge to analyze and interpret complex data, particularly to uncover patterns, trends, and insights from large datasets. Data scientists often build predictive models, design experiments, and communicate results through visualizations and reports. In academia, data scientists often collaborate with faculty members on academic research projects, applying data analysis techniques to disciplines like social science, biology, or education. However, data scientists also often work in industry to help companies to effectively derive insights from corporate data and support business decisions. Political representation in Germany is facilitated by the "German Data Science Society" [gds25]. Data science supports systematic approaches to exploratory analysis, prediction, and decision-making [Ge16]. It integrates quantitative techniques with domain knowledge to address challenges in fields like management, wireless networks, business, and public health [Ge16], while also bridging traditional statistics with computational and theoretical innovations to enhance reproducibility, interpretability, and ethical data use [Do17]. In Germany, data science is institutionally supported by organizations such as the German Data Science Society [Ge16], and conceptualized through multiple, complementary frameworks. Steinmann and Drechsler describe it as the ability to extract insights from complex, high-dimensional data, highlighting its critical role in data-intensive research [SD21]. The German Data Forum frames data science within a broader infrastructure that emphasizes valid and reliable data, robust management, and quality assurance practices [ed11]. Additionally, Hörner et al. position data science as a component of interdisciplinary doctoral education, emphasizing the importance of complementary competencies such as data literacy and stewardship [Hö21].
- **Research Software Engineers (RSEs)** combine expertise in software development

with an understanding of research processes and academic goals [KI16]. They develop, maintain, and improve the software that underpins modern research, ensuring it is reliable, reproducible, sustainable, and fit for scientific purposes. RSEs may work within one of the increasing number of research software engineering teams that have been set up at universities and research organisations over the past decade, or they may be embedded within a research team. They may have a job title that officially recognises them as an RSE, or they may have a standard research or technical job title such as Research Assistant, Research Fellow, or Software Engineer. Regardless of their job title, RSEs share a set of core skills that are required to design and develop research software, understand the research environment, and ensure that they produce sustainable, maintainable code that supports reproducible research outputs, following the Findability, Accessibility, Interoperability and Reusability (FAIR) principles [GAB+24]. RSEs have organised themselves in various national societies, in Germany this is de-RSE e.V [derse25].

- **Research Infrastructure Engineers / High Performance Computing (HPC) Engineers** support the running of research computing infrastructure that is vital to support the demands of the modern research lifecycle. While people have undertaken such roles for many years, they have often been embedded within or been a part of wider system administration or management roles. With the expansion of digital research and the associated computing infrastructure, we are increasingly seeing dedicated HPC and research computing infrastructure-focused roles that are based within teams or groups dedicated to supporting research computing. These individuals design, deploy and maintain research computing clusters and provide software and systems expertise to help researchers undertake their computations in an efficient and effective manner.

While there is overlap between all these fields, we can summarise that RSEs focus on the creation of software in order to facilitate research using their Software Engineering (SE) skills. Research Data Managers focus on the data that is generated and/or used during the research process and provide infrastructure for better provisioning of the data. Data Scientists focus on gaining insight from the data that is generated somewhere. Research Infrastructure Engineers focus on provisioning and running the computing platforms that software and data experts often use to support their work.

We focus our comparison in this paper on these two groups for three reasons:

- Within the digital technical professionals space, RSEs and Data Scientists are the two groups whose skills are most software focused. In this context, there can be significant crossovers in the skills developed and used, even though the focus of the two roles can be quite different.
- Similar to RSE, Data Science (DS) is a cross-cutting field that is applicable across a wide range of research domains. We believe it is important to understand how the application of competencies attributed to these roles may differ across domains.

- Despite the crossover between DS and RSE roles, we observe that the communities representing these roles are quite disconnected or independent, perhaps because of the different communities through which these roles have developed - this is something we are keen to understand in more detail.

The major computing organizations ACM and IEEE published a joint recommended computing curriculum which already includes data science: the Computing Curriculum 2020 [AI20] already lists DS as one of the special cases for computing competences. Whilst DS is mentioned, research software engineering was absent.

There are existing bodies of knowledge (BOKs) for both software engineering [IE14] and data science [ED17a], each offering a comprehensive mapping of knowledge areas among themselves and in relation to the ACM computing curriculum. However, the connection between competencies and the corresponding knowledge canon remains complex and nuanced. Some competencies are almost the equivalent to a knowledge-based learning goal whereas others integrate several knowledge areas with active application skills. For the intent of this paper the additional mapping of knowledge areas to competency would be out of scope. However, the idea of outlining the fields and mapping corresponding aspects can be transferred analogously. The DS-BOK maps data science not only to software engineering but also to (research) data management (RDM). This is interesting as RDM is another intersecting cycle.

(AND WHAT DOES THIS PAPER CONTRIBUTE? SUMMARIZE KEY INSIGHTS/FINDINGS HERE.) Our key contributions in this manuscript can be summarised as follows. We compared research software engineering and data science through the lens of the research cycle, examining how each discipline contributes at various stages. In doing so, we identified significant areas of overlap-such as the reliance on programming and analytical skills-as well as key differences in focus, methodology, and project outcomes. By devising a common language, we aim to highlight the shared competencies and curricular elements, fostering greater alignment and mutual understanding between the two fields.

The paper is structured as follows. . . .

2 RSE and DS embeddings in the Research Cycle

Both RSE and DS can be conceptualized as a cross-cutting concern in many disciplines. However, the definition and relevance of these issues can be generalized based on the function they fulfill in the research cycle Figure 1.

There are different research processes depending on the discipline and the research question. However, [De21a] showed that most of the research processes contain the following phases:

1. conceptualization (developing research questions, concepts)
2. design (developing the tools, instruments and concrete process models)

3. implementation (executing the experiment, study)
4. analysis & interpretation
5. dissemination (publishing, distributing, peer-review)
6. reflexion and improvements

For example, in the case of the field of learning technologies, the design phase often consists of extensive software development of different tools for learning. In this situation developing complex tools for learning can be considered as research software engineering. The analysis of what learners can gain from using these technologies can be conceived as an educational research in its own right. This example highlights the differences of scale in both the weight of the different process phases (here: phase 2) and the relevance of research engineering. It also shows a situation where research software engineering clearly differs from data science. A typical data science background would not enable researchers to build full-stack software that solves inefficiencies or hard-to-teach problems in education.

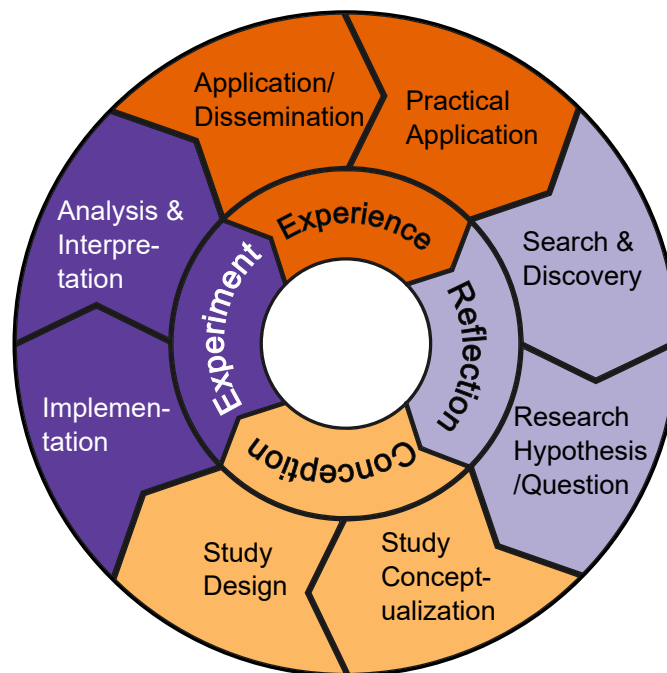


Fig. 1: The research cycle based on [Wi09] integrates the typical research process with the learning process.

In the second example a GPT-like attention model is trained to classify data gained from the James-Webb telescope. Due to vast amounts of data and the continuous stream of new data research software engineering is needed to implement a pipeline for data cleaning, data warehousing and in-time analysis. In this case, the analysis & interpretation phase (4) has much more relevance. Another point of this is example is that data science competencies such

as vectorization of algorithms, statistical analysis, machine learning etc. are interconnected with competencies from software engineering such as software architectures, software project management, and database programming. In this case the distinction between DS competencies and RSE competencies is very fluid.

The main argument behind these examples is that data science and research software engineering have a lot in common in terms of software development for science but show major differences where they are placed in the research process. This reframes the questions like “how much programming is in DS” or “how much engineering is in RSE” to a more structured approach of which cross-cutting functions exist in the research cycle that require computational means and which functions are part of the core identity of data scientists or research software engineers.

As a working hypothesis based on the example above and experience in the field we assume the following: [H1] RSE focuses more on concept development (if the research is computationally heavy), design, implementation and dissemination, i.e. phases 1,2,3,6 whereas DS focuses more on analysis, interpretation and dissemination (phases 4,5,6). Moreover, a second hypothesis would be that RSE often plays a role in shaping the context of the research [H2a], such as integrating projects with similar concerns, open source development and institutional needs. In contrast, data science is exclusively embedded in the research [H2b].

The focal point of the following chapter will revisit existing ideas for DS and RSE curricula and map the competences outlined there to the phases in the research process. This should give the abstract discussion above empirical grounding and can be used to test the hypothesis.

3 Method

In the following the contents of [Ge21] is parsed as the most current examples of data science curricula in the German research context. The contents are inspected for obvious links to the research process and interpreted if no explicit connections are made. Further sources for data science curriculum is the output of the Edison Project [ED17b] and the OpenDS4all Project [OD20].

For the RSE side the contents of [GAB+24] are used as a basis as well as the current state of the RSE-Curriculums project [De21b]. For the RSE competencies the RSE community has developed short codes. These are attached in the glossary in the attached repository [th25].

In addition, relevant competences from the research data management field [Pe25] are included as they intersect with both DS and RSE.

In terms of methodology it should be noted that this approach follows a community-driven consensus building. It should not be mistaken for a review study with measurable intersubjectivity based on instruments like PRISMA [Pa21].

The lists of extracted DS and RSE competency clusters can be found in [th25]. We also plan to publish a full table of RSE competencies (and possible DS competencies) sorted by clusters. In terms of this top-level discussion the competency clusters are sufficient to compare the differences with regard to the research cycle.

4 Discussion

H1 could not be confirmed in the strict sense. Although the compiled competency clusters show that there is a stronger focus on certain stages of the research process for both RSE and DS competencies, both can be interpreted more generally to encompass all stages of the research cycle. Moreover, there are some competency description that seem very similar such as the focus on the research cycle for RSE and the data science lifecycle for DS. In terms of methodology simply comparing the existing mentions of competencies should not be regarded as the best possible proxy to the actual distribution in the field. A survey study asking practitioners and researchers where in the process they would place DS and RSE would yield far more convincing results. Still, self-evaluation might also be biased depending on the identity of the people working in the respective fields.

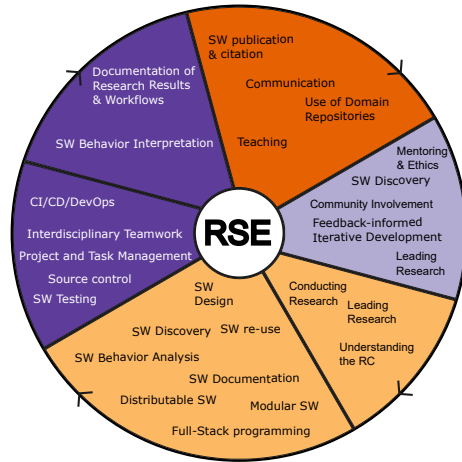


Fig. 2: RSE Competences in the Research Cycle

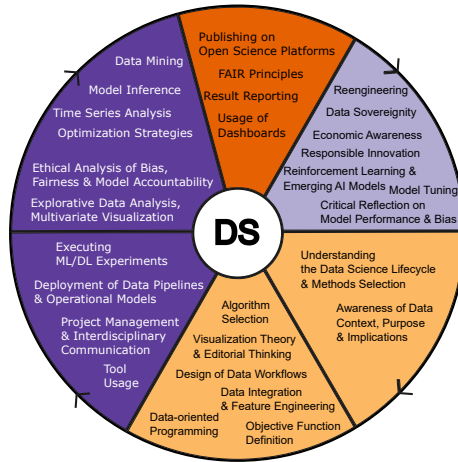


Fig. 3: DS Competences in the Research Cycle

The most clear differences between DS and RSE are found in the design stage and the analysis stage. The design stage holds most of the competency clusters the RSE community defined. The DS counterpart is very general and many competencies listed there could in fact be construed as RSE-competencies that are imported for more complex cases. In contrast, the analysis stage is more connected to DS. This can be explained by the historic challenges software development faces in terms of clear-cut evaluation but also by the distribution of labor: if the RSE-job ends with the developed software and the core experiment or study uses the software as a tool, the analysis part is then handed over to the respective field specialists.

Another reason for the design focus of RSE is the limited resources available. It is very time-consuming to both evaluate the impact of technology and also to evaluate the technology itself and its impact on the study. Comprehensive methods like Directed Acyclic Graph Modellings (DAGs) or Instrumental Variables try to tackle these nested evaluation issues but have not found widespread use. For that reason, the evaluation often concludes with the evaluation of the design part with instruments like the Technology Acceptance Model (TAM) or usability scales.

H2 also had to be rejected: in fact, DS seems to contain more aspects outside the research cycle than RSE. Even though the core analysis of data component is very embedded in research, DS has a lot of institutional, political and legal challenges. Research Data Management (RDM) could be named as the most prominent of these. Due to the strong overlap of non-research related competencies, a joint list competency clusters was compiled that lists the competences that are not research cycle related (also in the repository [th25]).

The long list of transversal competencies begs the question if there are also technical competences that overlap. Even though this was not the focus of the analysis, these can be easily spotted by investigating the shift of data analysis to artificial intelligence based methods. Training, fine-tuning and mainstreaming large language models requires more and more computing power, stable infrastructure and network components. On the other hand, CPU-based software-engineering becomes less demanding and also profits from AI-generated algorithm and code development. However, not all software engineering boils down to the current AI-hype. In summary, there is no clear way of generalizing whether DS or RSE need more and deeper understanding of computer science.

5 Conclusion

(planned/potential future work on this) We have given a short and non-exhaustive introduction into the profiles of the RSE, the Data Scientist, and the Research Data Manager. We have sketched their historical Development in order to enable practitioners from the other field participate in the discussion. The research cycle has served us as a useful backdrop on which to compare data science and research software engineering. While we found differences we have also seen commonalities. We hope that future work delineates the differences better in order to sharpen the profile. Also it will be beneficial for the curriculum development of both fields to understand their commonalities in order to share academic resources. We hope that this contribution can serve as an entry for more discussion about RSE in the Data Science community, and about the Data Science profile in the RSE community and thereby raising the awareness about each other in the communities. Among the broader trends in our world, we expect that a collaboration between the communities can be beneficial to harness the AI trend in a manner that is beneficial for public science.

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